

CSTE 5131

Machine Learning

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April 2025

Courtesy: Professor Qigang Gao
Dalhousie University



Cluster Analysis

- Clustering problem overview
- Data types for clustering
- K-means method
- Hierarchical methods
- Summary



Concepts of Clustering DM

- Goal: To find accurate clusters from data, i.e. groups of data objects representing different object concepts
 - Similar to one another within the same cluster
 - Dissimilar to the objects in other clusters
 - E.g., Customer groups, News discussion groups, Twits topic groups, Web page theme groups, etc.
- Clustering DM:
 - Methods/algorithms of dividing dataset into groups based on **similarity measure** (based on generic criteria)
 - No need of labeled training data (**unsupervised learning**)

Concepts of Clustering DM

- Find customer groups based on (age, credit, income)

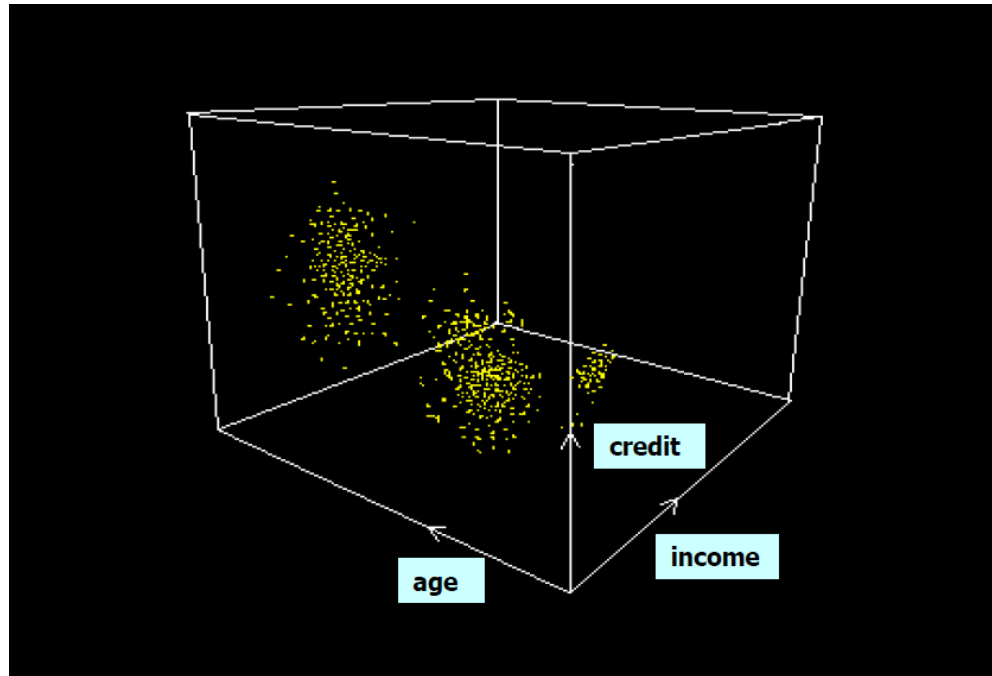
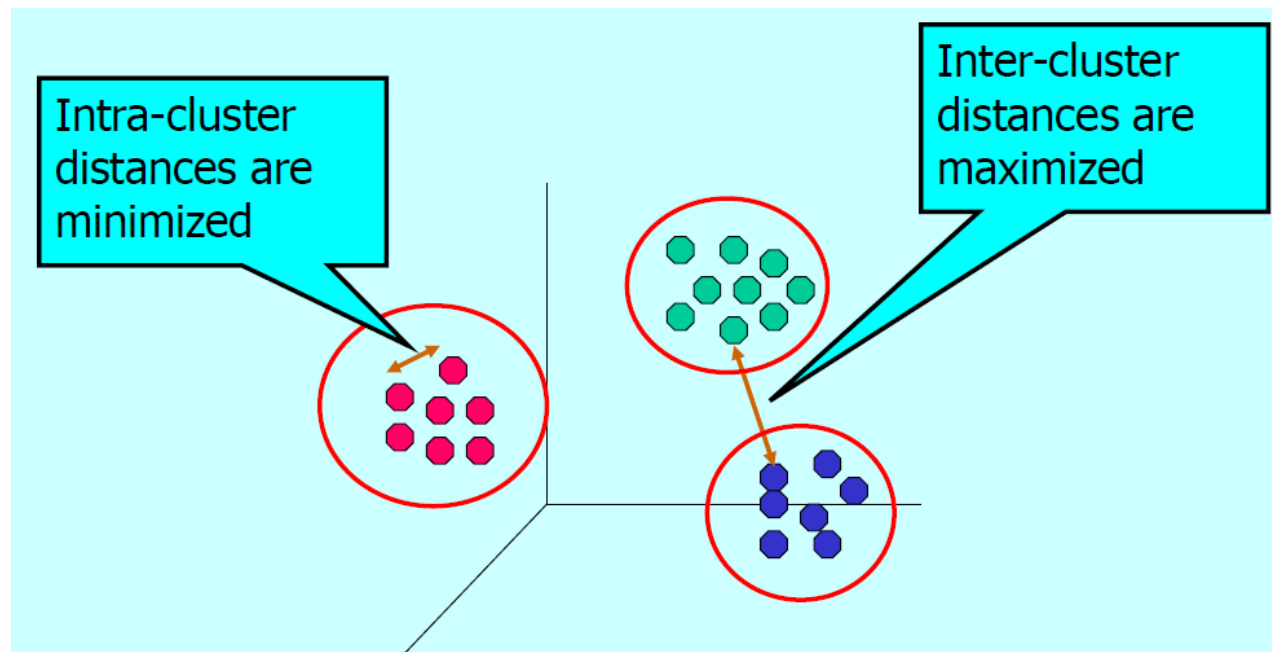


Illustration: Cluster Analysis

- Finding groups of objects which are similar in a group and dissimilar from different groups measured by distance criteria



What are the differences between
Clustering and Classification?



Clustering vs. Classification

- **Classification –Supervised Learning on a specific target concept**
 - Training data, i.e. samples with known answers of the target concept are required.
 - What happens when concepts/answers are hidden?
- **Clustering –Unsupervised Learning with no specific target concept**
 - No teacher/training set, which requires that the learner form and evaluate concepts on its own.
 - Clustering may be also viewed as unsupervised generic classification: the target is the data set itself, based on multiple attributes, there are no predefined classes.

Typical Usages

- When DM objective is less precise

- Clustering mining differs from other data mining techniques in that its objective is generally far less precise than the objectives of other predictive (e.g. classification) and descriptive (e.g. association) mining.

- We are always being told to "look at the big picture". But the fact is, sometimes the big picture is too confusing to be understood, so we may need to apply **"divide and conquer"** strategy to analyze the data.

- E.g., Customer behaviors analysis, customer profile DM, etc.

- As a stand-alone tool to get insight into hidden data concepts, and distribution to the concepts

- As a preprocessing step for other algorithms



Example: Video store data clustering for customer group analysis

Background: This project utilized data originating from a video outlet in Yarmouth, Nova Scotia containing transactional and customer information for a summer of 6 months. This database consists of approximately 10,000 customer records, approximately 50,000 fully detailed transactions, and information on the items being rented and sold. The **objective** is to find customer groups for improving marketing practices.

An instance of the DB (after data preprocessing):

CustID	nRentals	Revenue	nLate	LateChg	ReserveCount	Adult	Child	NewRel
49163	29	\$58.99	1	\$2.47	19	1	9	19
49164	14	\$31.22	10	\$37.70	0	3	7	4
49165	0	\$0.00	0	\$0.00	0	0	0	0
49166	22	\$123.86	7	\$22.23	6	0	1	15

E.g. The result when choosing $k=5$ with unit weights:

The dialog box titled "Automatic Cluster Detection" contains a "Clustering Weights" section with eight input fields, all set to 1: Rentals, Reservations, Revenue, Adult Rentals, Late, Child Rentals, Late (\$), and New Releases. To the right, the "Number Of Clusters" is set to 5. A "Cluster!" button is located below the weights. At the bottom is a table with 10 columns: ID, Rentals, Revenue, Late, Reserve, Adult, Child, wRelease, and Count. The table contains 5 rows of data. The first row (ID 0) has a "Reserve" value of 404 and a "Count" of 1, both circled in red. The last four rows (IDs 1-4) are enclosed in a red rectangular box. A red arrow points from the asterisk in the text below to the "Count" column of the first row.

ID	Rentals	Revenue	Late	Reserve	Adult	Child	wRelease	Count
0	6	19.34	0	404	3		1	1
1	2	7.1647	0	1	1	1	1	2837
2	35	105.4577	5	7	11	5	17	615
3	9	29.7416	1	2	3	3	5	2025
4	0	0.0072	0	1	4	1	1	5913

* An anomaly since it contains only one data instance but that one customer has reserved 404 movies. This would seem to indicate a problem that occurred during the initial phases of the data summarization and coding that led to this cluster.

E.g., Analytical Interpretation

Cluster	Description	# Of Customers
4	"Zero" cluster – customers that either <u>never or very rarely</u> purchase/rent from the store (in this period)	5913
1 *	Customers that <u>seldom rent</u> from the store (Averaging 2-3 transactions in a 6 month period)	2837
<u>3</u> **	Customers with <u>more frequent transactions</u> (Ave: 9 transactions in the 6 month period)	2025
<u>2</u> ***	Customers with an <u>extremely high frequency</u> of transactions (Ave: 35) and sales volume. (This cluster would form the business' most valuable customers)	615
Total Customers		11,390



Example: Clustering newsgroup emails for supporting information retrieval application

A sample of Email:

Return-path: <LUS.MD.GOOJU@lpch.stanford.edu>
Received: from DIRECTORY-DAEMON by SYSWRK.UCIS.DAL.CA (PMDF V4.3-13 #6307)
id <01JEAINJT3NK001RMV@SYSWRK.UCIS.DAL.CA>; Mon, 02 Aug 1999 12:26:06 -0300
Received: from 171.65.56.137 by SYSWRK.UCIS.DAL.CA (PMDF V4.3-13 #6307)
id <01JEAINBMNQ8001NW4@SYSWRK.UCIS.DAL.CA>; Mon, 02 Aug 1999 12:25:55 -0300
Received: from meditech.Stanford.EDU by LPCH.Stanford.Edu with SMTP
(Microsoft Exchange Internet Mail Service Version 5.0.1458.49)
id P98FYL31; Mon, 2 Aug 1999 08:25:29 -0700
Date: Mon, 02 Aug 1999 08:27:16 -0700
From: LUS.MD.GOOJU@lpch.stanford.edu
To: PEDIATRIC-PAIN@ac.dal.ca
Message-id: <990803527.LUS475898@lpch.stanford.edu>
Content-transfer-encoding: 7BIT
To: PEDIATRIC-PAIN@ac.dal.ca
From: GOOD,JULIE
Date: August 2, 1999 08:27am
Our nurses use the Wong-Baker FACES on a ten point scale, with each of the
faces divided by an extra point. We are not sure if this affects the validity
or reliability of the measure - but it facilitates a smooth transition to the
pure numeric scale when the patients are older.
-Julie J. Good, M.D.
Pediatric Pain Management Fellow
Packard Children's Hospital at Stanford

Define Clustering

- Given a dataset D with n tuples: $D=\{t_1, t_2, \dots, t_n\}$, and a parameter k , the **Clustering Problem** is to find a mapping $f : D \Rightarrow \{1, \dots, k\}$ where each t_i is assigned to one cluster K_j , $1 \leq j \leq k$
- A **Cluster**, K_j , contains precisely those tuples mapped to it (with the highest similarity)
- A general similarity measure method: **Euclidean distance**

What is good Clustering?

- A good clustering method will produce high quality clusters with
 - high intra-class similarity
 - low inter-class similarity
- The quality of a clustering result depends on both the similarity measure used by the method and its implementation.



General Criteria of Clustering (Book Sec: 10.1.2)

- Scalability
- Ability to deal with different types of attributes
- Discovery of clusters with arbitrary shape
- Minimal requirements for domain knowledge to determine input parameters
- Ability to deal with noise and outliers
- Insensitive to order of input records
- High dimensionality
- Incorporation of user-specified constraints
- Interpretability and usability



Data Types for Clustering

- Textbook: Section 2.4
- The records in a data set need to be mapped into points (position vector) in a multidimensional space
- How to handle features with **non-numerical domains**?
 - Attributes can be
 - non-numerical, such as binary, nominal (categorical), ordinal variables.
 - Some numeric variables such as rankings do not have the right behavior to properly be treated as components of a position vector
 - Records for purchases, phone calls, airplane trips, addresses, and many other things that have no obvious connection to the dots in a cluster diagram.

Data Structures for Clustering

- With a dataset of n objects:
 - **Data matrix (object-by-variable structure)**
 - o Two modes: n objects by p variables
 - o A form of relational table or $n \times p$ matrix
 - **Dissimilarity matrix (object-by-object structure)**
 - o One mode: n objects by n objects, and
 $d(i, j) = d(j, i)$, $d(i, i) = 0$
 - o A form of $n \times n$ table, stores available differences for all pairs of n objects

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

- Relationship of the two matrices:
 - Many clustering algorithms operate on dissimilarity matrix directly, in which data matrix is used for serving data preparation process.

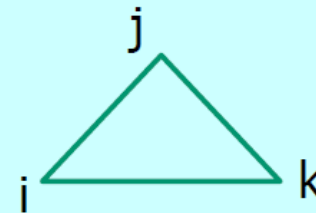
Dissimilarity Between Objects

- Euclidean distance:

$$d(i, j) = \sqrt{(|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \dots + |x_{ip} - x_{jp}|^2)}$$

- Properties

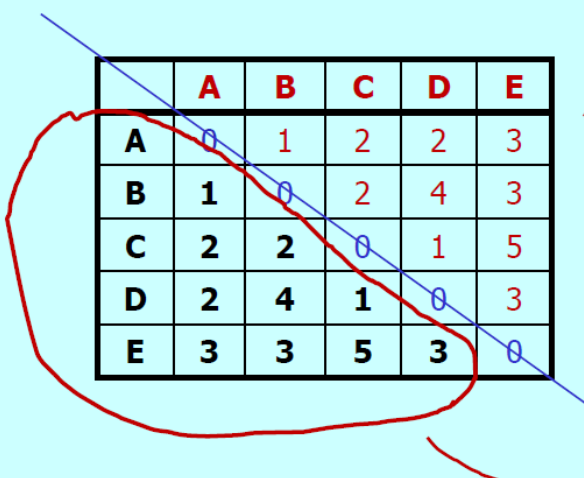
- $d(i, j) \geq 0$
- $d(i, i) = 0$
- $d(i, j) = d(j, i)$
- $d(i, j) \leq d(i, k) + d(k, j)$



- Also, one can use weighted distance, or other dissimilarity measures

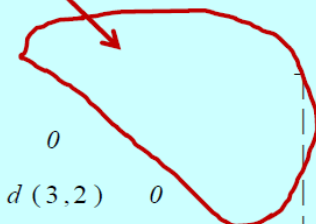
Dissimilarity Between Objects

E.g. A dissimilarity matrix with available distances of objects {A,B,C,D,E}



	A	B	C	D	E
A	0	1	2	2	3
B	1	0	2	4	3
C	2	2	0	1	5
D	2	4	1	0	3
E	3	3	5	3	0

Redundant: $d(i,j) = d(j,i)$

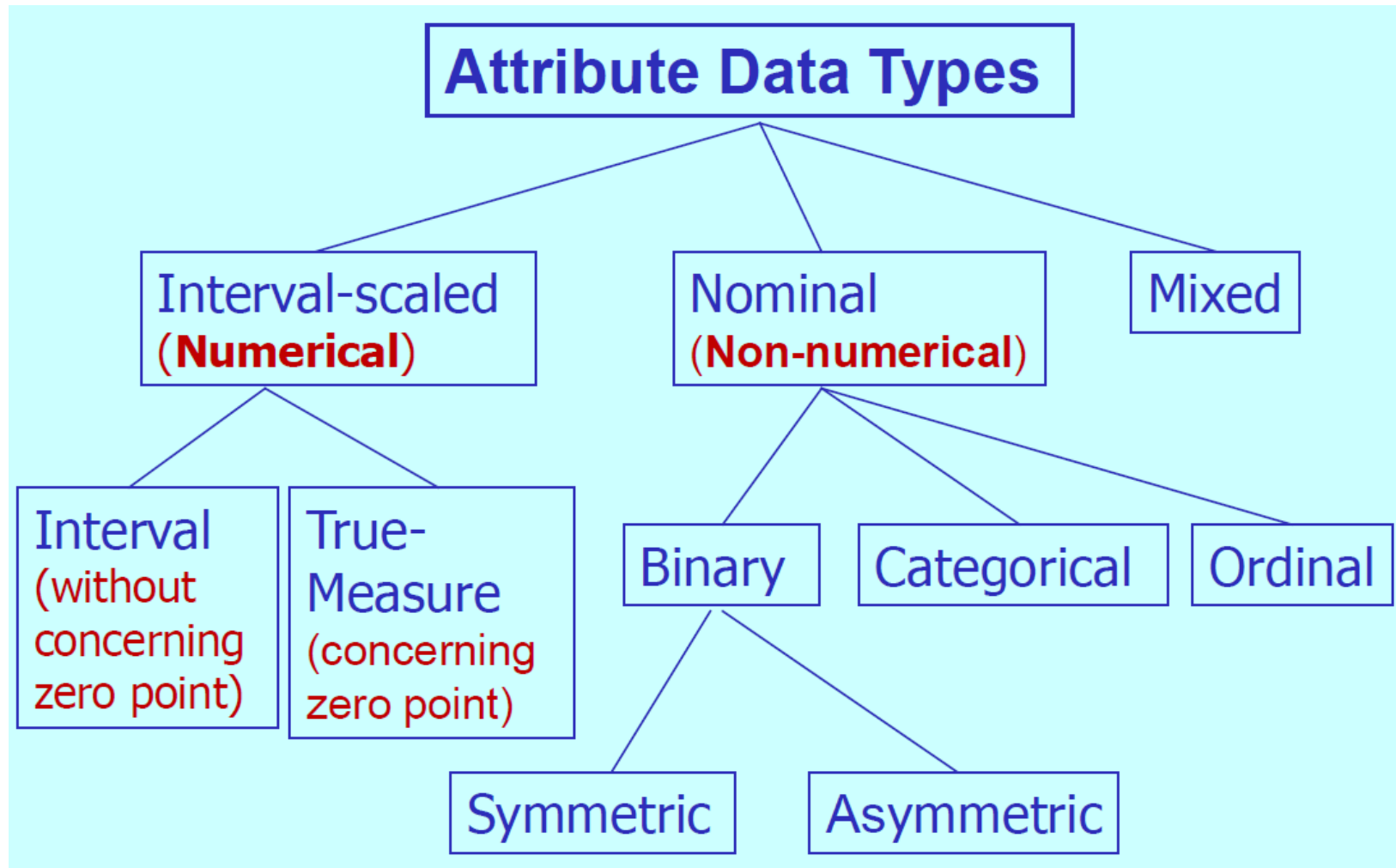

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$



Main Attribute Types for Clustering

- Numeric (Interval-scaled or ratio scaled) variables
- Binary variables
- Categorical variables
- Ordinal variables
- Variables of mixed types

Section 2.1



Interval-valued variables

- Interval Variables:

- Describe the distance between two measure values without concerning a zero point, such as temperature, etc.

- E.g., A day's highest temperature: 20 C° and lowest temperature is 10 C° , the highest difference = $20 - 10 = 10\text{ C}^\circ$.

- 0 C° doesn't mean no temperature (zero doesn't indicate absence of the variable)

- Ratio has no meaning for interval variables

- E.g., Is a temperature 20 C° warmer twice than 10 C° ?

- No. but it is true to say it is 10 C° warmer.

- True Measure Variables:

- An interval variables with a zero point (zero represents the complete absence of the variable)

- The ratio of two values of the variable is meaningful

- E.g., A distance 1000 km is twice far of a distance 500 km .

- Distance, volume, age, etc. are examples of true measures.



Proximity Measures for Binary Variables

- How to measure dissimilarity between objects with binary variables?
 - E.g., Hospital patient database

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

- Two values of binary variables may not have equal value (i.e., importance) for an application
 - E.g., In the given medical application, HIV tested as positive will have more impact to the overall outcome. If two variables of such have positive (or “1s”), the outcome would be considered more significant than of two “0s”.



Proximity Measures for Binary Variables

- Dissimilarity and similarity measures for objects described by either *symmetric* or *asymmetric binary attributes*.
- A **contingency table** for p binary variables of object i and object j :
 - q is the number of attributes that equal 1 for both objects i and j
 - r is the number of attributes that equal 1 for object i but equal 0 for object j
 - s is the number of attributes that equal 0 for object i but equal 1 for object j ,
 - and t is the number of attributes that equal 0 for both objects i and j .
 - The total number of attributes is p , where

$$p = q + r + s + t$$

		Object j		
		1	0	sum
Object i	1	q	r	$q + r$
	0	s	t	$s + t$
	sum	$q + s$	$r + t$	p



Proximity Measures for Binary Variables

- **Symmetric Binary Dissimilarity:**

- each state is equally valuable for symmetric binary attributes
- Dissimilarity that is based on symmetric binary attributes is called **symmetric binary dissimilarity**

- The dissimilarity between i and j is:

$$d(i,j) = \frac{r+s}{q+r+s+t}$$

- **Asymmetric binary dissimilarity:**

- For asymmetric binary attributes, the two states are not equally important,
- The dissimilarity based on these attributes is called **asymmetric binary dissimilarity**, where the number of negative matches, t , is considered unimportant

$$d(i,j) = \frac{r+s}{q+r+s}$$

- **Asymmetric binary similarity:**

$$sim(i,j) = \frac{q}{q+r+s} = 1 - d(i,j)$$

- The coefficient $sim(i, j)$ is called the **Jaccard coefficient**



Proximity Measures for Binary Variables

- Example: Dissimilarity between binary attributes:

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	1	0	1	0	0	0
Mary	F	1	0	1	0	1	0
Jim	M	1	1	0	0	0	0

Asymmetric binary variables

- The distance between Jack and Jim, $d(\text{jack}, \text{jim}) = 1+1/1+1+1 = 0.67$
- The distance between Jack and Mary, $d(\text{jack}, \text{mary}) = 0+1/2+0+1 = 0.33$



Proximity Measures for Nominal Attributes

(Sec:2.4.2)

- The dissimilarity between two objects i and j can be computed based on the ratio of mismatches:

$$d(i,j) = \frac{p-m}{p}$$

where m is the number of *matches* (i.e., the number of attributes for which i and j are in the same state), and p is the total number of attributes describing the objects.

Table 2.2 A Sample Data Table Containing Attributes of Mixed Type

Object Identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}$$



Dissimilarity of Numeric Data (Sec: 2.4.4)

- Commonly used distance measures for numeric attributes:
 - *Euclidean*,
 - *Manhattan*,
 - and *Minkowski distances*.

- The Euclidean distance between objects i and j is defined as,

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{ip} - x_{jp})^2}.$$

- Manhattan distance,

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{ip} - x_{jp}|.$$

- Minkowski distance is a generalization of the Euclidean and Manhattan distances. It is defined as,

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}.$$

Where, h is a real number such that $h \geq 1$



Dissimilarity of Numeric Data (Sec: 2.4.4)

- The **supremum distance** (also referred to as L_{max} , L_{∞} **norm** and as the **Chebyshev distance**) is a generalization of the Minkowski distance for $h \rightarrow \infty$.
- To compute it, we find the attribute f that gives the maximum difference in values between the two objects.

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|.$$

- Let $x_1 = (1, 2)$ and $x_2 = (3, 5)$ represent two objects

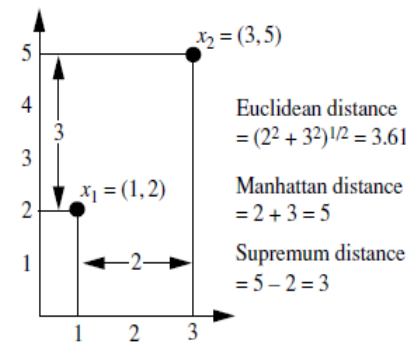


Figure 2.23 Euclidean, Manhattan, and supremum distances between two objects.

Proximity Measures for Ordinal Attributes (Sec: 2.4.5)

- The values of an ordinal attribute have a meaningful order or ranking about them
- The dissimilarity computation with respect to attribute, f involves the following steps:
 - Replace each x_{if} by its corresponding rank, $r_{if} \in \{1, \dots, M_f\}$
 - map the range of each attribute onto $[0.0, 1.0]$ so that each attribute has equal weight

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}.$$

- Compute dissimilarity using a distance measure method

Proximity Measures for Ordinal Attributes (Sec: 2.4.5)

- Example: dissimilarity between ordinal attributes

Table 2.2 A Sample Data Table Containing Attributes of Mixed Type

Object Identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

$$\begin{bmatrix} 0 & & & \\ 1.0 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix}$$

- There are three states for test-2: fair, good, excellent, that is $M_f = 3$
- The four objects are assigned the ranks 3, 1, 2, and 3, respectively.
- Mapping: rank 1 to 0.0, rank 2 to 0.5, and rank 3 to 1.0.
- Dissimilarity matrix (using Euclidean distance)



Basic Clustering Methods

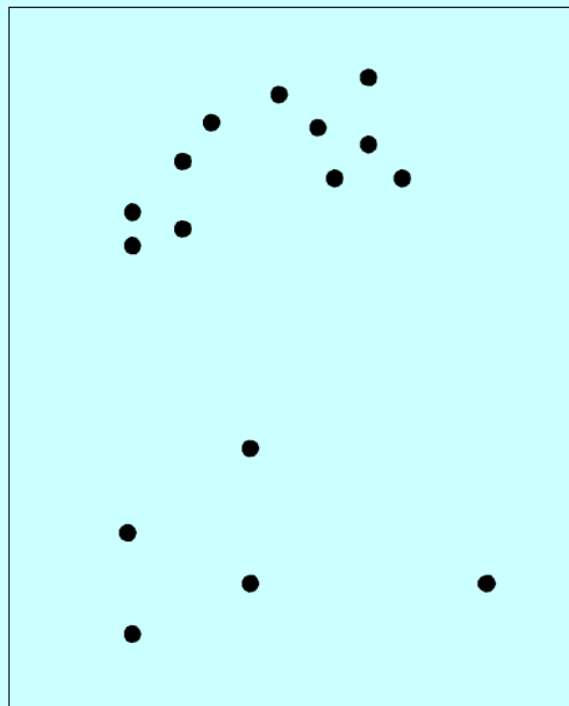
- Partitioning methods (k-means, k-medoids)
- Hierarchical methods
 - Agglomerative
 - Divisive
- Density-based methods (E.g., Density-based spatial clustering of applications with noise (DBSCAN))
- Grid-based



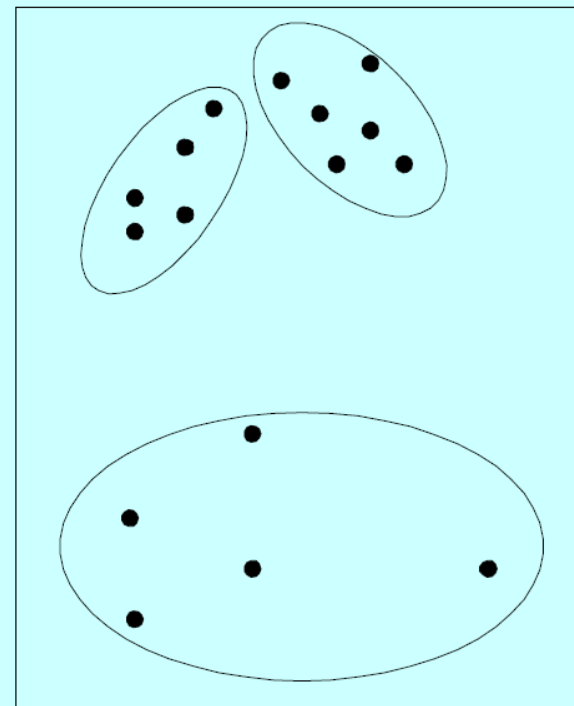
Basic Clustering Methods

Method	General Characteristics
Partitioning methods	– Find mutually exclusive clusters of spherical shape – Distance-based – May use mean or medoid (etc.) to represent cluster center – Effective for small- to medium-size data sets
Hierarchical methods	– Clustering is a hierarchical decomposition (i.e., multiple levels) – Cannot correct erroneous merges or splits – May incorporate other techniques like microclustering or consider object “linkages”
Density-based methods	– Can find arbitrarily shaped clusters – Clusters are dense regions of objects in space that are separated by low-density regions – Cluster density: Each point must have a minimum number of points within its “neighborhood” – May filter out outliers
Grid-based methods	– Use a multiresolution grid data structure – Fast processing time (typically independent of the number of data objects, yet dependent on grid size)

Partitional Clustering

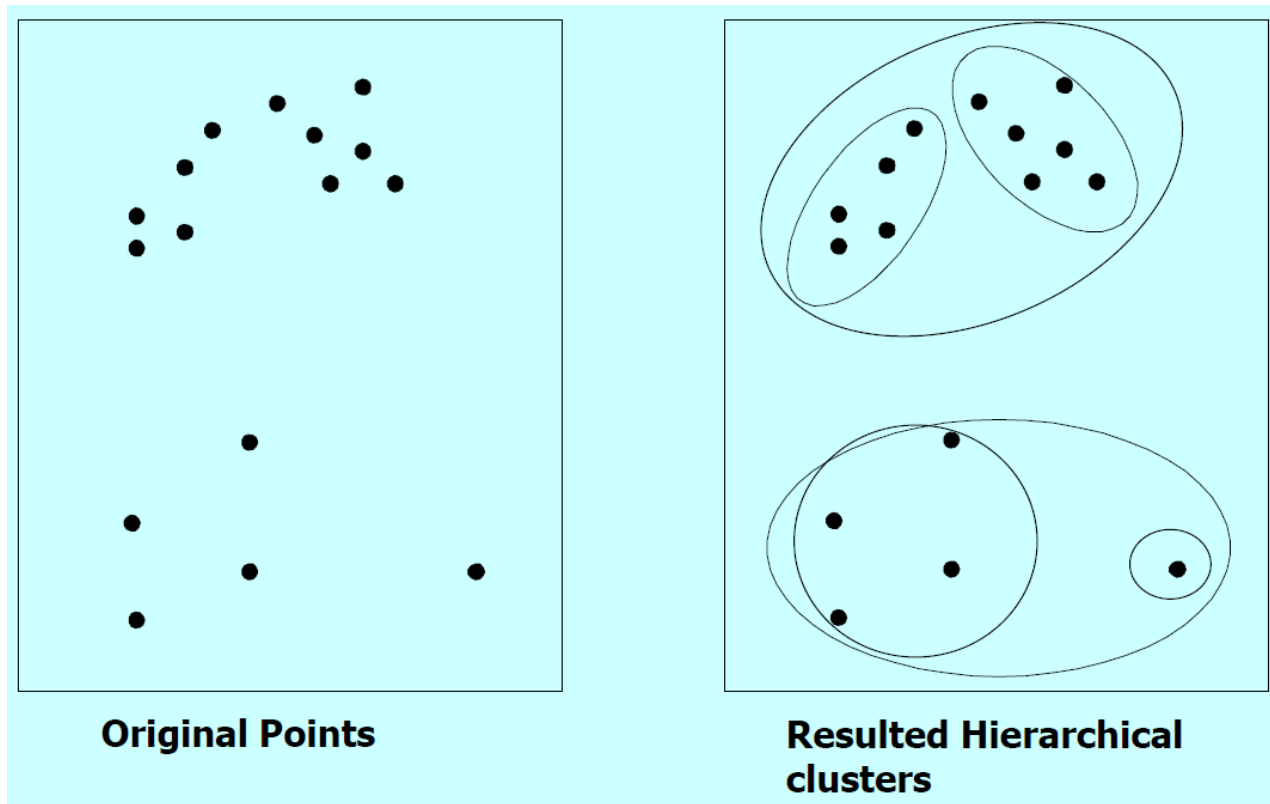


Original Points



Resulted flat clusters

Hierarchical Clustering



Partition-based Approach



- It is a **single level** clustering technique: data partition without hierarchy
- It creates clusters in **one stage** as opposed to several stages
- Since only one set of clusters is output, the user normally has to input the desired number of clusters, k

K-means Clustering

- K-means (MacQueen'67):
A centroid-based technique
- In general, the algorithm works as follows:

1. Let training set, $T = x_1, \dots, x_n$ and $x_i \in R^m$ i.e. each tuple represents a point in an m dimensional feature space. Parameter K is used as the number of clusters. Randomly Choose K objects in sequence from training set as the initial cluster centers i.e. $c_1, \dots, c_k$. Let $C_1, \dots, C_k$ represents K partitions or clusters of training data.

2. Determine the distance of each object to the centroid by using Euclidian distance.

$$dist(x_i, c_k) = \sqrt{\sum_{s=1}^m (x_{is} - c_{ks})^2}$$

where $i = 1, \dots, n$ and $k = 1, \dots, K$

3. Group the objects based on minimum distance from cluster center to each object and update cluster membership $C_1, \dots, C_k$ based on that distance. Grouping data set into K cluster to minimize the within cluster sum of square distance which is defined as:

$$\sum_{k=1}^K \sum_{j \in C_k} ||X_j - C_k||^2$$

4. Recalculate the cluster centers by averaging the points belong to that cluster:

$$c_k = \frac{1}{|C_k|} \sum_{j \in C_k} x_j$$

where $k = 1, \dots, K$

5. Repeat step 2 to step 4 until convergence.



K-means clustering

- Example: Given 1) a 1D data set: {2, 4, 10, 12, 3, 20, 30, 11, 25},
2) $K = 2$.
 - Methods:
 - Randomly assign initial cluster center: $c_1 = 3, c_2 = 4$
 - Calculate new centers, then relabel objects into K clusters until converge (i.e., new means/centers have no change):
1. $K1=\{2, 3\}, K2=\{4, 10, 12, 20, 30, 11, 25\}, c_1=2.5, c_2=16$
 2. $K1=\{2, 3, 4\}, K2=\{10, 12, 20, 30, 11, 25\}, c_1=3, c_2=18$
 3. $K1=\{2, 3, 4, 10\}, K2=\{12, 20, 30, 11, 25\}, c_1=4.75, c_2=19.6$
 4. $K1=\{2, 3, 4, 10, 11, 12\}, K2=\{20, 30, 25\}, c_1=7, c_2=25$
 5. **$K1$** ={2, 3, 4, 10, 11, 12}, **$K2$** ={20, 30, 25}, $c_1=7, c_2=25$ (No more change), iteration stops.
- Example (2D data set): Explore the link below
<https://people.revoledu.com/kardi/tutorial/kMean/NumericalExample.htm>

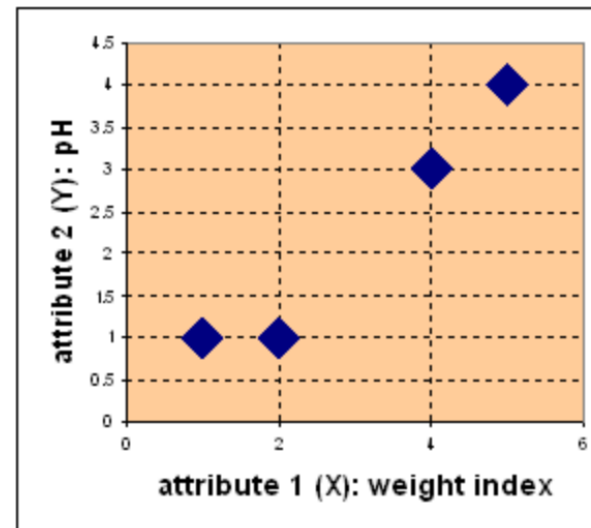


K-means clustering

- Example (2D data set):

Suppose we have several objects (4 types of medicines) and each object have two attributes or features as shown in table below. Our goal is to group these objects into K=2 group of medicine based on the two features (pH and weight index).

Object	Attribute 1	Attribute 2
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

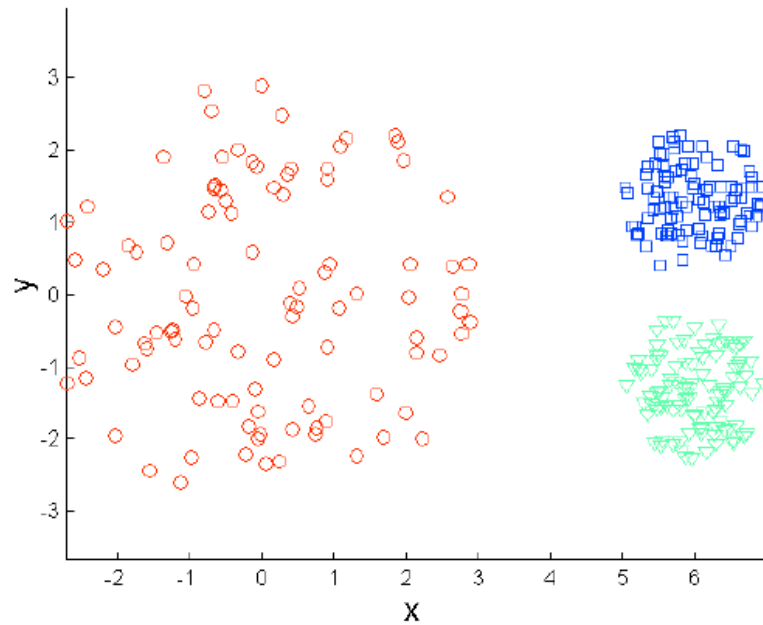


K-means clustering

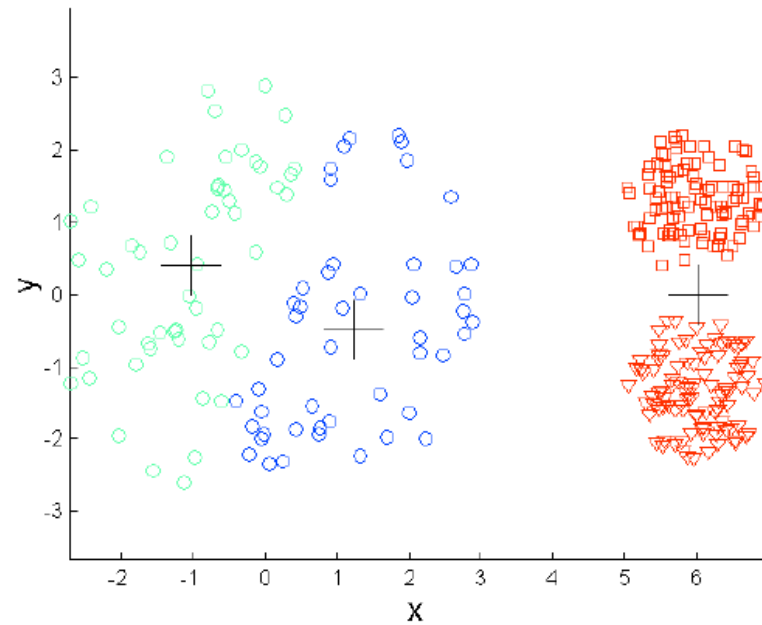


- Evaluating K-means Clusters
 - Sum of Squared Error (SSE)
 - Reduce SSE
- Properties of K-means
- Limitations of K-means
 - Selection of k and initial cluster centers.
 - Unable to handle datasets with different types of attributes.
 - Final solution after convergence might not be the global minimum.
 - Does not work well with non-globular clusters.
- Impact of Outliers on Clustering

Limitations of K-means (different density)

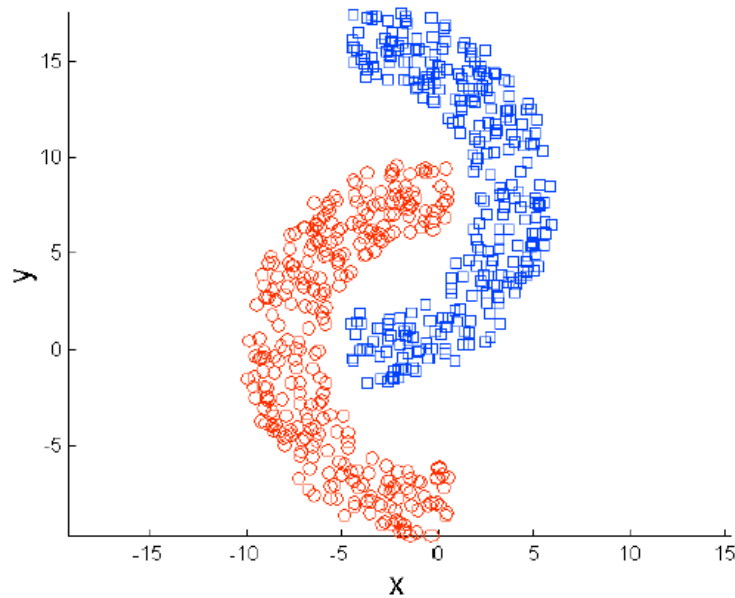


Original Points

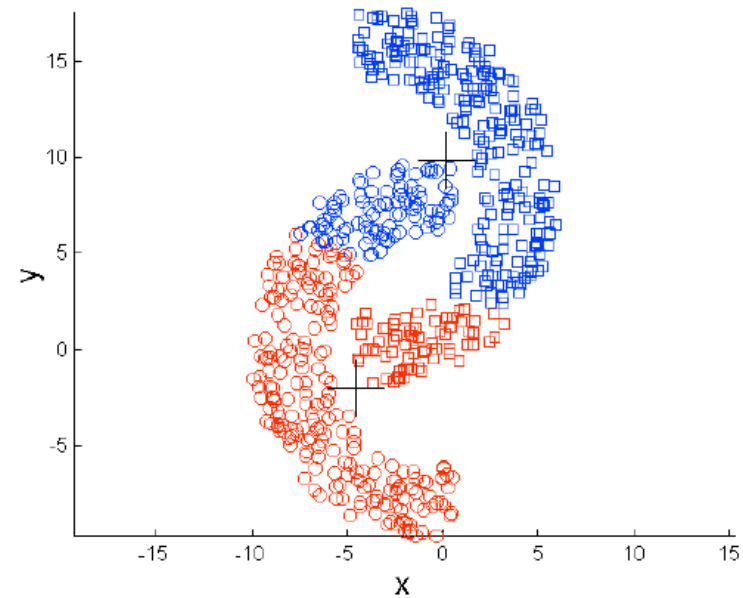


K-means (3 Clusters)

Limitations of K-means (with non-globular shape)



Original Points



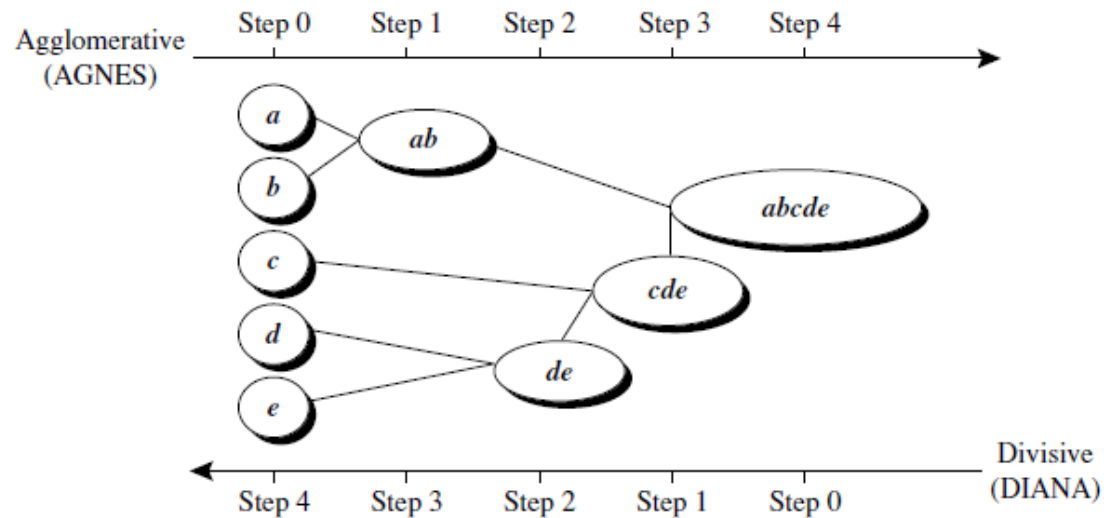
K-means (2 Clusters)

Hierarchical methods



- Clusters are formed in levels
 - actually, creating sets of clusters at each level
- Agglomerative
 - Initially each data point in its own cluster
 - Iteratively clusters are merged together
 - Bottom Up
- Divisive
 - Initially all items in one cluster
 - Large clusters are successively divided
 - Top Down

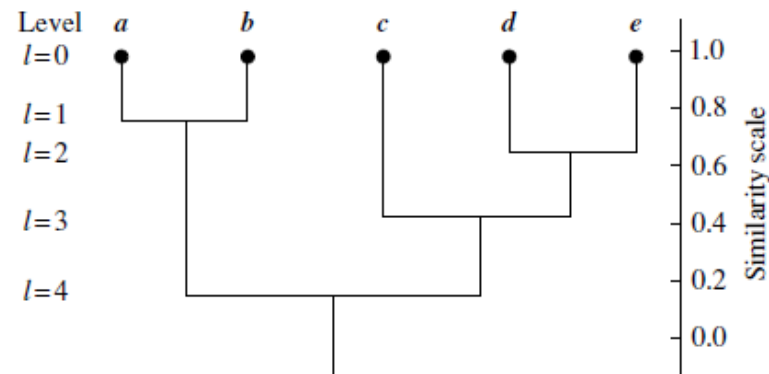
Hierarchical methods



Agglomerative and divisive hierarchical clustering on data objects $\{a, b, c, d, e\}$

- Use distance matrix as clustering criteria. This method does not require the number of clusters k as an input, but needs a termination condition

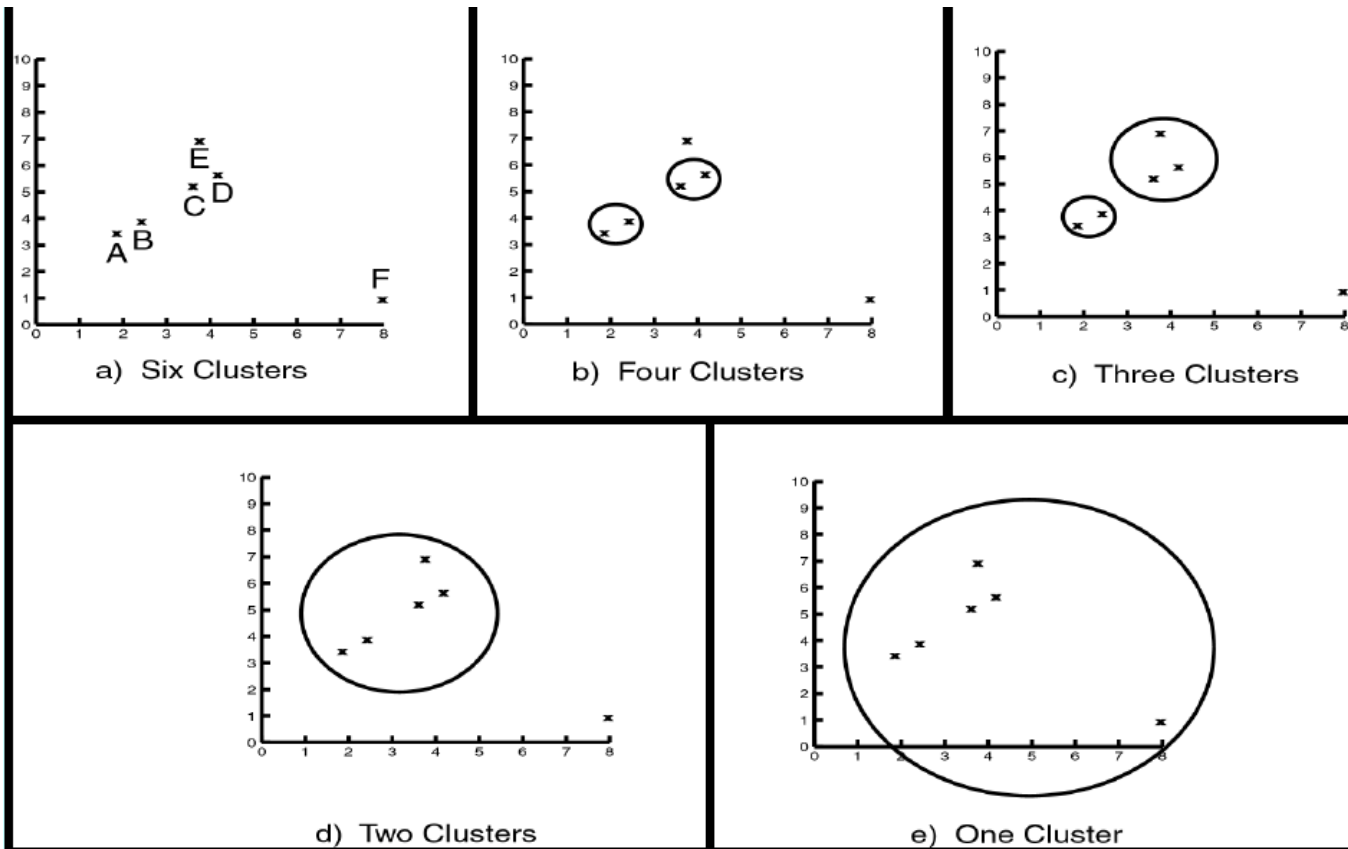
Hierarchical methods



Dendrogram representation for hierarchical clustering of data objects $\{a, b, c, d, e\}$

- **Dendrogram:** a tree data structure which illustrates hierarchical clustering techniques
- Each level shows clusters for that level.
 - Leaf - individual clusters
 - Root - one cluster

Hierarchical methods

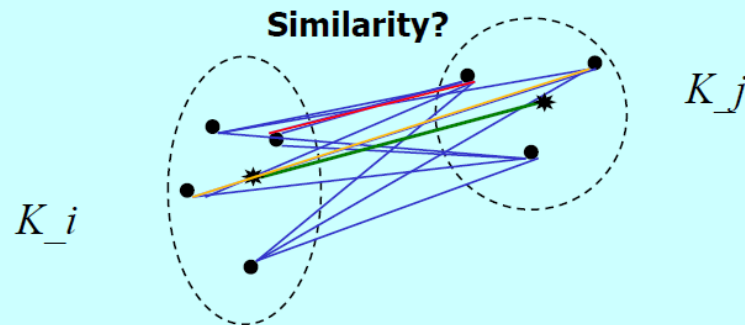


Dendrogram
Representation
?

Distance between Clusters

- Hierarchical algorithms require that the distance between clusters (rather than elements) be determined. There are several standard alternatives:

- **Single Link (MIN)**: smallest distance between points.
- **Complete Link (MAX)**: largest distance between points.
- **Average Link**: average distance between points.
- **Centroid Link**: distance between centroids.



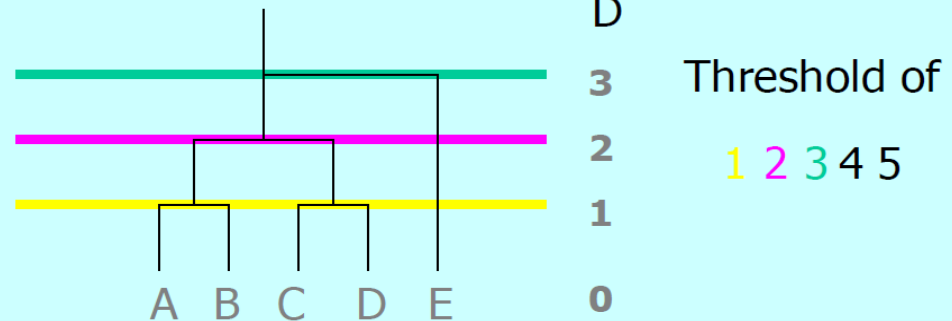
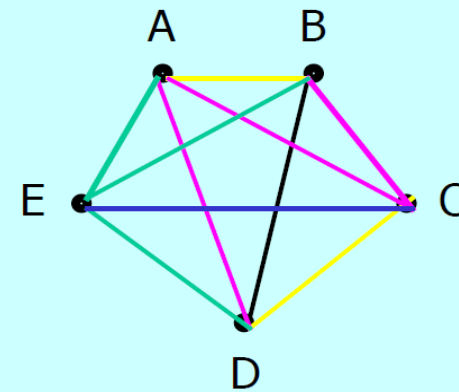
See also
Section: 10.3.2



Example: Agglomerative clustering

- Use the **Single-Link** method and the **dissimilarity matrix**

	A	B	C	D	E
A	0				
B	1	0			
C	2	2	0		
D	2	4	1	0	
E	3	3	5	3	0





Homework!

- Dissimilarity for Attributes of Mixed Types (Section: 2.4.6)
- Read class note, and chapter 2 (mainly Section 2.4), chapter 10 (mainly Section 10.1, 10.2, 10.3 (until 10.3.2))

Thank you

